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Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2021

Subject: Analysis and Design of Concrete Structure

Code: 101501

Time: 03 Hours

Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.
(v) Use of IS 456-2000 and other relevant IS codes is permitted.
(vi) Assume the necessary relevant data with justification, if required.

Q.1 Choose the correct answer of the following (any seven) : **[14]**

(a) In the limit state method, balanced design of a reinforced concrete beam gives the
- (i) smallest concrete section and maximum area of reinforcement
- (ii) largest concrete section and maximum area of reinforcement
- (iii) smallest concrete section and minimum area of reinforcement
- (iv) largest concrete section and minimum area of reinforcement

(b) The bond strength between steel and concrete is due to
- (i) friction
- (ii) adhesion
- (iii) both friction and adhesion
- (iv) None of the above

(c) The depth of footing for an isolated column is governed by
- (i) maximum bending moment
- (ii) shear force
- (iii) punching shear
- (iv) All of the above

(d) To minimise the effect of differential settlement, the area of a footing should be designed for
- (i) dead load only
- (ii) dead load + live load
- (iii) dead load + fraction of live load
- (iv) live load + fraction of dead load

(e) Maximum percentage of reinforcement in case of slabs is limited to
- (i) 2
- (ii) 4
- (iii) 6
- (iv) 8

(f) The main reason for providing number of reinforcing bars at a support in a simply supported beam is to resist
- (i) compressive stress
- (ii) shear stress
- (iii) bond stress
- (iv) tensile stress

(g) When shear stress exceeds the permissible limit in a slab, then it is reduced by
- (i) increasing the depth
- (ii) providing shear reinforcement
- (iii) using high strength steel
- (iv) using thinner bars but more in number

(h) Minimum pitch of transverse reinforcement in a column is
- (i) the least lateral dimension of the member
- (ii) sixteen times the smallest diameter of longitudinal reinforcement bar to be tied
- (iii) forty-eight times the diameter of transverse reinforcement
- (iv) Lesser of the above three values

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(i) For a simply supported beam of span 15 m, the minimum effective depth to satisfy the vertical deflection limits should be

(i) 60 cm
(ii) 75 cm
(iii) 90 cm
(iv) more than 1 m

(j) According to IS: 456, minimum slenderness ratio for a short column is

(i) less than 12
(ii) less than 18
(iii) between 18 and 24
(iv) more than 24

Q.2 Explain the following terms: **[14]**

(a) Limit state
(b) Characteristic strength
(c) Characteristic load
(d) Modulus of elasticity of concrete
(e) Stress-strain curve for concrete
(f) Creep and shrinkage
(g) Acceptance criteria of concrete

Q.3 Solve both questions : **[14]**

(a) Discuss the salient features of limit state method. Discuss partial safety factors for loads under limit state of collapse and limit state of serviceability.

(b) What are the different limit states of limit state design philosophy over other design philosophies?

Q.4 Solve this question : **[14]**

A three-span continuous beam carries a UDL of 35 kN/m under service conditions. The clear span of beam is 4.5 m with support width as 300 mm. Design the beam using M25 grade concrete and Fe500 steel.

Q.5 Solve this question : **[14]**

Design a slab for a room 3.0 m x 4.0 m clear in size, if the superimposed load is 3.0 kN/m^2. Consider for both the cases with and without corners, free to lift and the edges are simply supported. Use M25 mix and Fe500 grade steel.

Q.6 Solve this question : **[14]**

Design the reinforcements of a rectangular beam simply supported over a clear span of 4.3 m by limit state method, if the superimposed load is 75 kN at mid-span and each support width is 450 mm. Neglect the self-weight of beam. Use M25 mix and Fe500 grade steel.

Q.7 Solve this question : **[14]**

Using limit state method, design a reinforced concrete column, 400 mm square, to carry an ultimate load of 1000 kN at an eccentricity of 160 mm. Use M25 mix and Fe415 grade steel.

Q.8 Solve both questions : **[14]**

(a) Explain the codal recommendations for the effective span of the stair slabs with different support conditions of the loadings.

(b) Design a dog-legged staircase located in a stair hall measuring 2.75 m x 4.5 m for a public building. The story of floor-to-floor height is 3.3 m. The service live load on stairs is 4.0 kN/m^2 with the weight of finishes of 0.5 kN/m^2. Adopt riser and tread of 150 mm and 250 mm, respectively. The grades of concrete and steel to be used are M25 and Fe500, respectively.

Q.9 Solve both questions : **[14]**

(a) Write the procedure to design an isolated rectangular footing to carry axial loads and moments.

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(b) Design a footing for a 350 mm square column reinforced with 8-20 mm Fe415 grade bars. One side of the footing is restricted to 1.5 m. The gross bearing capacity of the soil is 65 kN/m^2. Assume column concrete as M25 and footing concrete as M20.

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